

Linear Programming (LP)

General function

$$\text{objective function } \max f(x_1, x_2, x_3, \dots, x_n) = \sum_{j=1}^n c_j x_j$$

$$s.t. \begin{cases} g_1(x_1, x_2, x_3, \dots, x_n) \leq b_1 \\ g_2(x_1, x_2, x_3, \dots, x_n) \leq b_2 \\ \dots \\ g_m(x_1, x_2, x_3, \dots, x_n) \leq b_m \end{cases}$$

$$g_i(x_1, x_2, x_3, \dots, x_n) = \sum_{j=1}^n a_{ij} x_j, \text{ where } i = 1, 2, 3, \dots, m$$

Linear Programming

$$\text{objective function } \max \left(\min \right) y = (\text{a linear combination of independent variables})$$

$s.t.$ all the constraints

notice that for some programming, the objective function can only be calculated as the maximum, hence need to [transform the minimum function into a maximum function](#), by adding a negative sign

e.g. A dairy factory produces two products: A1 and A2, by using milk as raw materials. A bucket of milk can be used to produce 3kg of product A1 within 12 hours in workshop p or 4kg of product A2 within 8 hours in workshop q . Assume that all products A1 and A2 can be sold, with profits of 24\$/kg and 16\$/kg respectively. The total working hour is 480 hours a day, the production ability of workshop p is 100kg per day and the production ability of workshop q can be considered as infinite. The supply of milk is 50 buckets per day. Determine a production plan, in order to reach maximum profit.

assume that x, y buckets of milk are used to produce product A1 and A2 respectively

$$\max k = 72x + 64y$$

$$s.t. \begin{cases} x + y \leq 50 \\ 12x + 8y \leq 480 \\ 3x \leq 100 \\ x \geq 0, y \geq 0 \end{cases}$$

